B8 · Cold drink

Model derivation, step by step

darkpyxlz · SYPT 2027

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The current model is two coupled lumped systems — the drink, described by its temperature T_d, and the bath, described by its total enthalpy $\mathcal{H}$ — joined by one heat flow, $UA(T_d - T_b)$. This document derives both equations from conservation laws and closes them with phase equilibrium and convection correlations. No numbers are substituted; every result is left in symbols. Step 13 collects every result into the closed equation set.

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Scope and conventions

What "current model" means. The 24 Oct salted-ice-bath model: lumped drink, enthalpy-based bath with a regime map (liquid / liquidus / eutectic plateau / saturated / solid), Bodnar liquidus, box leak and stirrer power, with the $\Delta T^{1/4}$ convection law as a switchable option. The buffered eutectic bath (COMSOL v1) falls out as a special case.

How to read it. Each step states its assumptions, derives, and ends in a boxed **Result**. Step 13 collects every Result into the closed equation set; Step 14 gives the special cases with closed-form solutions.

Conventions.

- Bath and drink temperatures are in °C, because the Bodnar liquidus and the enthalpy reference are defined at 0 °C. Absolute temperature (K) appears only in the phase-equilibrium derivation (Step 6), written $\mathcal{T}$.
- Salt content w is a mass fraction (0–1). Bodnar's fit is in wt %, so it is written with $W = 100w$.
- A heat flow is positive in the direction its subscript names: $\dot{Q}_{db}$ flows drink → bath, $\dot{Q}_{\text{leak}}$ flows environment → bath.
- Enthalpy reference state: pure liquid water and solid NaCl, both at 0 °C, have zero enthalpy.

Nomenclature

Source column: **M** measured on this rig, **T** tabulated / literature, **D** derived here, **S** state variable.

Symbol	Meaning	Unit	Src
T_d	drink temperature (well-mixed core)	°C	S
T_b	bath temperature	°C	D (from $\mathcal{H}$)
T_{env}	ambient air outside the box	°C	M
θ	drink–bath difference, $T_d - T_b$	K	D
$m_d,\ c_d$	drink mass, specific heat	kg, $\mathrm{J\,kg^{-1}K^{-1}}$	M, T
$m_c,\ c_c$	container mass, specific heat	kg, $\mathrm{J\,kg^{-1}K^{-1}}$	M, T
C_d	drink + container heat capacity	$\mathrm{J\,K^{-1}}$	D
$k_d, \rho_d, \nu, \alpha, \beta$	drink conductivity, density, kinematic viscosity, diffusivity, expansion coefficient	SI	T
$r_i,\ r_o,\ H$	container inner radius, outer radius, height	m	M
$t_w,\ k_w$	wall thickness, wall conductivity	m, $\mathrm{W\,m^{-1}K^{-1}}$	M, T
$A_i,\ A_o$	inner and outer wetted areas	$\mathrm{m^2}$	M
$h_i,\ h_o$	drink-side and bath-side film coefficients	$\mathrm{W\,m^{-2}K^{-1}}$	D/M
R_i, R_w, R_o, R_c	inner film, wall, outer film, contact (label) resistances	$\mathrm{K\,W^{-1}}$	D
$U,\ UA$	overall coefficient, conductance	$\mathrm{W\,m^{-2}K^{-1}}$, $\mathrm{W\,K^{-1}}$	M (fit)
Bi, Nu, Ra, Re, Pr	Biot, Nusselt, Rayleigh, Reynolds, Prandtl numbers	—	D
$\mathcal{H}$	total bath enthalpy	J	S
$m_W,\ m_S$	total water (ice + liquid), total NaCl	kg	M
$m_i,\ m_l,\ m_h$	ice, brine, undissolved solid salt	kg	D
$w,\ \bar{w}$	brine salt fraction; overall fraction $m_S/(m_W + m_S)$	—	D

Symbol	Meaning	Unit	Src
$T_{\mathrm{liq}}(w)$	liquidus (ice–brine coexistence) temperature	°C	T/M
$w_{\mathrm{sat}}(T)$	salt solubility curve	—	T
$T_E,\ w_E$	eutectic temperature and composition	°C, —	T
b_1, b_2, b_3	Bodnar (1993) coefficients: 1.78, -0.0442, 5.572×10^{-4}	—	T
c_i, c_b, c_h, c_w	specific heats of ice, brine, solid NaCl, water	$\mathrm{J\,kg^{-1}K^{-1}}$	T
L_f	latent heat of fusion of ice at 0 °C	$\mathrm{J\,kg^{-1}}$	T
Δh_s	enthalpy of solution per kg NaCl (>0 endothermic)	$\mathrm{J\,kg^{-1}}$	T
$L_{\mathrm{eff}},\ C_{\mathrm{eff}}$	effective latent heat; effective bath heat capacity	$\mathrm{J\,kg^{-1}}$, $\mathrm{J\,K^{-1}}$	D
Λ_E	enthalpy to form 1 kg of eutectic brine from ice + salt	$\mathrm{J\,kg^{-1}}$	D
UA_{box}	box (EPS) conductance	$\mathrm{W\,K^{-1}}$	M
P_{stir}	stirrer power dissipated in the bath	W	M
$\mathcal{T},\ \mathcal{T}_0$	absolute temperature; 273.15 K	K	—
$M_{\mathrm{H_2O}},\ M_{\mathrm{NaCl}}$	molar masses	$\mathrm{kg\,mol^{-1}}$	T
Δh_{fus}	molar enthalpy of fusion of ice	$\mathrm{J\,mol^{-1}}$	T
$a_w,\ \varphi,\ \nu_i$	water activity, osmotic coefficient, ions per formula unit (2)	—	T
$b,\ K_f$	molality; cryoscopic constant	$\mathrm{mol\,kg^{-1}}$, $\mathrm{K\,kg\,mol^{-1}}$	D
$\tau,\ \tau_0,\ t^*$	time constants	s	D
$n,\ a$	exponent and prefactor in $\dot{Q} = a\theta^n$	—, $\mathrm{W\,K^{-n}}$	D/M
T_{fd}	freezing point of the drink	°C	T/M

Step 0 — System, control volumes and assumptions

The apparatus splits into two systems with heat capacity — the drink and the bath — and two treated as pure resistances — the container wall and the box. Everything outside the box is a reservoir at T_{env}.

Control volume	Contents	State	Exchanges heat with
Drink (D)	liquid + container shell	T_d	bath, through the wall
Bath (B)	ice, brine, undissolved NaCl	$\mathcal{H}$	drink; environment via box; stirrer work
Container wall	Al or PET shell, label	—	resistance only
Box	EPS walls + lid	—	resistance only

Thermal circuit.

$$\underbrace{T_d}_{C_d} \xrightarrow{R_i} T_{wi} \xrightarrow{R_c+R_w} T_{wo} \xrightarrow{R_o} \underbrace{T_b}_{\mathcal{H}} \xleftarrow{1/UA_{\mathrm{box}}} T_{\mathrm{env}}, \qquad P_{\mathrm{stir}} \to \text{bath}$$

Assumptions.

A1 Drink well-mixed. One temperature T_d describes the liquid core; all variation sits in thin boundary layers. Tested in Step 3.

A2 Wall and tank have negligible heat capacity, or theirs is added to the system they touch (container into C_d; tank glass adds $C_t T_b$ to $\mathcal{H}$).

A3 Bath well-mixed and at local phase equilibrium at every instant.

A4 Quasi-steady resistances. Boundary layers respond faster than T_d or T_b change.

A5 Constant pressure, lid on. No evaporation, condensation or frost; radiation absorbed

into UA_{box}.

A6 Constant properties, except h_i and β (Step 5).

A7 No hydrohalite, no metastability. The salt-rich solid is anhydrous NaCl, the eutectic is the equilibrium T_E, nothing supercools.

A8 Drink fully immersed to a marked waterline: wetted area fixed.

A9 Ideal sensors. Probe lag and offsets are measurement corrections, not model terms.

Step 1 — First law for the drink

The drink plus container is a closed system at constant pressure doing no work, so $\mathrm{d}H_D/\mathrm{d}t$ = heat in $= -\dot{Q}_{db}$. With constant specific heats (A6) and one temperature (A1),

$$H_D = (m_d c_d + m_c c_c)(T_d - T_{\text{ref}}) \quad \Rightarrow \quad \frac{\mathrm{d}H_D}{\mathrm{d}t} = (m_d c_d + m_c c_c)\frac{\mathrm{d}T_d}{\mathrm{d}t} \equiv C_d \frac{\mathrm{d}T_d}{\mathrm{d}t}.$$

Result 1 — drink energy balance

$$C_d \frac{\mathrm{d}T_d}{\mathrm{d}t} = -\dot{Q}_{db}, \qquad C_d = m_d c_d + m_c c_c$$

m_d comes from weighing the drink full and empty, not from the external volume $\pi r_o^2 H$, which includes headspace.

Step 2 — Heat path drink → bath as series resistances

2.1 Each layer carries the same heat flow (A4):

$$\dot{Q}_{db} = \frac{T_d - T_{wi}}{R_i} = \frac{T_{wi} - T_{wo}}{R_w} = \frac{T_{wo} - T_b}{R_o}.$$

2.2 Film resistances from Newton's law, $\dot{Q} = hA\,\Delta T$:

$$R_i = \frac{1}{h_i A_i}, \qquad R_o = \frac{1}{h_o A_o}.$$

2.3 Wall resistance of a cylinder. Fourier's law through a shell of height H, $\dot{Q}$ the same at every r:

$$\dot{Q} = -k_w(2\pi r H)\frac{\mathrm{d}T}{\mathrm{d}r} \;\Rightarrow\; \int_{r_i}^{r_o}\frac{\mathrm{d}r}{r} = -\frac{2\pi k_w H}{\dot{Q}}\int_{T_{wi}}^{T_{wo}}\mathrm{d}T \;\Rightarrow\; R_w = \frac{\ln(r_o/r_i)}{2\pi k_w H}.$$

For a thin wall, $t_w = r_o - r_i \ll r_i$, $\ln(r_o/r_i) \approx t_w/r_i$ and $R_w \approx t_w/(k_w A)$. A label sleeve adds $R_c = R_c''/A$ in series.

2.4 Add the series. Writing each equality as a temperature drop and summing, the wall temperatures cancel:

$$T_d - T_b = \dot{Q}_{db}\,(R_i + R_c + R_w + R_o).$$

2.5 Faces in parallel. Side, base and (if submerged) top are parallel paths j:

$$UA = \sum_j \left(\frac{1}{h_{i,j} A_{i,j}} + R_{c,j} + R_{w,j} + \frac{1}{h_{o,j} A_{o,j}} \right)^{-1},$$

$$\frac{1}{U} = \frac{1}{h_i} + R_c'' + \frac{t_w}{k_w} + \frac{1}{h_o} \quad \text{(thin wall, } A_i \approx A_o \approx A\text{)}.$$

2.6 What the wall probe measures.

$$\frac{T_d - T_{wo}}{T_{wo} - T_b} = \frac{R_i + R_c + R_w}{R_o}.$$

Can versus bottle. For aluminium $t_w/k_w \ll 1/h$, so R_w drops out; PET keeps a finite R_w, which matters more once stirring shrinks R_o. Compare containers by U, not τ.

Result 2 — heat flow from drink to bath

$$\dot{Q}_{db} = UA\,(T_d - T_b), \qquad \frac{1}{UA} = R_i + R_c + R_w + R_o$$

Step 3 — Biot number and when lumping is allowed

A still drink fails the lumping test, so A1 holds only because natural convection mixes the core: the model is a mixed core plus a thin boundary layer, not a conducting solid.

3.1 The test.

$$\mathrm{Bi} = \frac{U_{\mathrm{ext}} L_c}{k_d}, \qquad \frac{1}{U_{\mathrm{ext}}} = R_c'' + \frac{t_w}{k_w} + \frac{1}{h_o}, \qquad L_c = \frac{V}{A} \text{ or } r_i.$$

Conduction-only lumping needs $\mathrm{Bi} < 0.1$. The often-quoted tiny Biot number for an aluminium can uses k_w and t_w: it is the *wall's* Biot number.

3.2 A conducting drink in a long cylinder with a fixed-temperature wall decays with

$$\tau_{\mathrm{cond}} = \frac{r_i^2}{\alpha\, j_{0,1}^2}, \qquad j_{0,1} = \text{first zero of } J_0.$$

3.3 The rescue. Buoyancy drives a boundary layer of thickness $\delta \sim H\,\mathrm{Ra}_H^{-1/4} \ll r_i$; its resistance is R_i, already in the circuit. Enhancement: $E = \tau_{\mathrm{cond}}/\tau_{\mathrm{measured}}$.

Result 3 — lumping criterion

A1 holds if $\delta \sim H\,\mathrm{Ra}_H^{-1/4} \ll r_i$ and $|T_{\mathrm{top}} - T_{\mathrm{bottom}}| \ll T_d - T_b$ (not $\mathrm{Bi} < 0.1$)

Step 4 — Fixed-bath solution

Combine Results 1 and 2 with T_b constant and $\theta = T_d - T_b$:

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\frac{\theta}{\tau}, \quad \tau \equiv \frac{C_d}{UA} \;\Rightarrow\; \int_{\theta_0}^{\theta} \frac{\mathrm{d}\theta'}{\theta'} = -\int_0^t \frac{\mathrm{d}t'}{\tau} \;\Rightarrow\; \ln\frac{\theta}{\theta_0} = -\frac{t}{\tau}.$$

4.4 Derived quantities.

$$t_{T^*} = \tau \ln \frac{T_{d0} - T_b}{T^* - T_b}, \qquad \left.\frac{\mathrm{d}T_d}{\mathrm{d}t}\right|_0 = -\frac{T_{d0} - T_b}{\tau},$$

$$\frac{t_{T^*}(T_{b1})}{t_{T^*}(T_{b2})} = \frac{\ln[(T_{d0}-T_{b1})/(T^*-T_{b1})]}{\ln[(T_{d0}-T_{b2})/(T^*-T_{b2})]}, \qquad \frac{\dot{T}_d(0)|_{T_{b1}}}{\dot{T}_d(0)|_{T_{b2}}} = \frac{T_{d0}-T_{b1}}{T_{d0}-T_{b2}}.$$

The two ratios differ: "how much faster" has two honest answers.

4.5 Geometry inside τ. With $C_d \approx \rho_d c_d V$: $\tau = (\rho_d c_d/U)\,(V/A)$.

Result 4 — fixed-bath cooling curve

$$T_d(t) = T_b + (T_{d0}-T_b)\,e^{-t/\tau}, \qquad \tau = \frac{C_d}{UA}$$

Step 5 — Convection closures and the power-law cooling curve

5.1 Drink side, laminar natural convection (length scale H, film drop $\Delta T_i = T_d - T_{wi}$):

$$\mathrm{Ra}_H = \frac{g\beta\,\Delta T_i H^3}{\nu\alpha}, \qquad \mathrm{Nu}_H = \frac{h_i H}{k_d} = C_i\,\mathrm{Ra}_H^{1/4} \;\Rightarrow\; h_i = C_i k_d \left(\frac{g\beta}{\nu\alpha H}\right)^{1/4} \Delta T_i^{1/4} \equiv K_i\,\Delta T_i^{1/4}.$$

5.2 Drink film dominant ($R_i \gg R_c + R_w + R_o$, so $\Delta T_i \approx \theta$): $\dot{Q}_{db} = K_i A_i\,\theta^{5/4} \equiv a\,\theta^{5/4}$.

5.3 Integrate with T_b fixed.

$$\int_{\theta_0}^{\theta} \theta'^{-5/4}\mathrm{d}\theta' = -\frac{a}{C_d}\int_0^t \mathrm{d}t' \;\Rightarrow\; \theta^{-1/4} = \theta_0^{-1/4} + \frac{at}{4C_d} \;\Rightarrow\; \theta(t) = \theta_0\left(1+\frac{t}{t^*}\right)^{-4},$$

$$t^* = \frac{4C_d}{a\,\theta_0^{1/4}} = 4\tau_0, \qquad \tau_0 = \frac{C_d}{h_i(\theta_0)A_i}.$$

5.4 General exponent ($\dot{Q} = a\theta^n$, $n \neq 1$):

$$\theta^{1-n} = \theta_0^{1-n} + (n-1)\frac{a}{C_d}t \;\Rightarrow\; \theta(t) = \theta_0\left[1+(n-1)t/\tau_0\right]^{-1/(n-1)}, \qquad \tau_0 = \frac{C_d}{a\theta_0^{n-1}}.$$

$n \to 1$ recovers Result 4.

5.5 Exponent test.

$$\ln\left(-\frac{\mathrm{d}T_d}{\mathrm{d}t}\right) = \ln\frac{a}{C_d} + n\ln(T_d - T_b).$$

5.6 Mixed resistances (implicit in $\dot{Q}_{db}$):

$$\dot{Q}_{db} = K_i A_i\left[\theta - \dot{Q}_{db}\,(R_c + R_w + R_o)\right]^{5/4}, \qquad 1 \le n_{\text{eff}} \le \tfrac{5}{4}.$$

5.7 Bath side.

$$\text{unstirred: } \mathrm{Nu}_o = C_o\left(\frac{g\,\Delta\rho\,H^3}{\rho_b\nu_b\alpha_b}\right)^{1/4}, \qquad \text{stirred: } \mathrm{Nu}_o = \frac{h_o D}{k_b} = C\,\mathrm{Re}^m\mathrm{Pr}^{1/3}, \quad \mathrm{Re} = \frac{u_\infty D}{\nu_b}.$$

5.8 Caveat — $\beta(T)$. Near the density maximum use the density difference directly:

$$\mathrm{Ra}_H = \frac{g\left[\rho_d(T_{wi}) - \rho_d(T_d)\right]H^3}{\rho_d\,\nu\,\alpha}.$$

Fit n only over the early stretch of a run.

Result 5 — convection closure and fixed-bath power law

$$h_i = K_i(T_d - T_{wi})^{1/4}, \qquad \dot{Q}_{db} = a\theta^n,\ 1 \le n \le \tfrac{5}{4}, \qquad \theta(t) = \theta_0\left(1 + \frac{t}{t^*}\right)^{-4} \quad (n = \tfrac{5}{4})$$

Step 6 — Phase equilibrium of the bath

6.1 Equilibrium condition.

$$\mu_{\text{ice}}(\mathcal{T}) = \mu_w^*(\mathcal{T}) + R\mathcal{T}\ln a_w \ \Rightarrow\ R\mathcal{T}\ln a_w = -\Delta g_{\text{fus}}(\mathcal{T}).$$

6.2 Gibbs–Helmholtz with $\Delta g_{\text{fus}}(\mathcal{T}_0) = 0$, Δh_{fus} constant:

$$\frac{\Delta g_{\text{fus}}}{\mathcal{T}} = \Delta h_{\text{fus}}\left(\frac{1}{\mathcal{T}} - \frac{1}{\mathcal{T}_0}\right) \ \Rightarrow\ \ln a_w = -\frac{\Delta h_{\text{fus}}}{R}\left(\frac{1}{\mathcal{T}} - \frac{1}{\mathcal{T}_0}\right).$$

6.3 Dilute, ideal limit → Blagden. $\ln a_w \approx -\nu_i b M_{\text{H}_2\text{O}}$, $\mathcal{T}\mathcal{T}_0 \approx \mathcal{T}_0^2$:

$$\Delta T_f = \underbrace{\frac{R\mathcal{T}_0^2 M_{\text{H}_2\text{O}}}{\Delta h_{\text{fus}}}}_{K_f}\nu_i b, \qquad b = \frac{w/M_{\text{NaCl}}}{1-w}; \qquad \frac{1}{\mathcal{T}_{\text{liq}}} = \frac{1}{\mathcal{T}_0} + \frac{R}{\Delta h_{\text{fus}}}\varphi(w)\,\nu_i\,b(w)\,M_{\text{H}_2\text{O}} \quad \text{(real brine)}.$$

6.4 Empirical liquidus (Bodnar 1993). With $\vartheta = -T_{\text{liq}}$ (°C), $W = 100w$:

$$W = b_1\vartheta + b_2\vartheta^2 + b_3\vartheta^3, \qquad \frac{\mathrm{d}T_{\text{liq}}}{\mathrm{d}w} = -\frac{100}{b_1 + 2b_2\vartheta + 3b_3\vartheta^2} < 0.$$

6.5 Gibbs phase rule at fixed pressure, $C = 2$: $F = 3 - P$.

Phases present	P	F	What is fixed
brine	1	2	nothing
ice + brine	2	1	$T = T_{\text{liq}}(w)$
brine + solid salt	2	1	$w = w_{\text{sat}}(T)$
ice + brine + solid salt	3	0	$T = T_E$, $w = w_E$
ice + solid salt	2	1	T free, below T_E

6.6 Eutectic and solubility.

$$T_{\text{liq}}(w_E) = T_E, \qquad w_{\text{sat}}(T_E) = w_E, \qquad w_{\text{sat}}(T) \approx w_E + (w_p - w_E)\frac{T - T_E}{T_p - T_E}.$$

Result 6 — phase constraints

ice + brine: $T_b = T_{\text{liq}}(w)$; brine + salt: $w = w_{\text{sat}}(T_b)$; all three: $T_b = T_E,\ w = w_E$

Step 7 — Bath mass balances

$$m_W = m_i + (1-w)m_l, \qquad m_S = w\,m_l + m_h, \qquad m_W + m_S = m_i + m_l + m_h.$$

With no solid salt ($m_h = 0$):

$$m_l = m_W + m_S - m_i, \qquad w = \frac{m_S}{m_W + m_S - m_i};$$
$$\frac{\mathrm{d}m_l}{\mathrm{d}m_m} = 1, \qquad \frac{\mathrm{d}w}{\mathrm{d}m_m} = -\frac{m_S}{m_l^2} = -\frac{w}{m_l} \quad (\mathrm{d}m_m = -\mathrm{d}m_i).$$

Whether $\bar{w} = m_S/(m_W + m_S)$ lies below or above w_E decides which phase runs out first on the plateau.

Result 7 — mass constraints

$$m_W = m_i + (1-w)m_l, \quad m_S = w\, m_l + m_h, \quad 0 \le m_i \le m_W, \quad 0 \le m_h \le m_S$$

Step 8 — Bath enthalpy and the bath energy equation

8.1 Specific enthalpies (reference: liquid water and solid NaCl at 0 °C; dilution enthalpy neglected):

$$h_{\mathrm{ice}} = c_i T - L_f, \qquad h_{\mathrm{salt}} = c_h T, \qquad h_{\mathrm{brine}} = c_b T + w\,\Delta h_s.$$

8.2 Total enthalpy.

$$\mathcal{H} = m_i(c_i T_b - L_f) + m_l(c_b T_b + w\Delta h_s) + m_h c_h T_b \quad [+ C_t T_b \text{ if the tank is included}].$$

8.3–8.4 First law and box leak.

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = \dot{Q}_{db} + \dot{Q}_{\mathrm{leak}} + P_{\mathrm{stir}}, \qquad \dot{Q}_{\mathrm{leak}} = U A_{\mathrm{box}}(T_{\mathrm{env}} - T_b).$$

8.5 Energy check. Adding Result 1, $\dot{Q}_{db}$ cancels: with the box sealed and no stirrer, $C_d T_d + \mathcal{H}$ is constant.

Result 8 — bath energy equation

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = UA(T_d - T_b) + UA_{\mathrm{box}}(T_{\mathrm{env}} - T_b) + P_{\mathrm{stir}}$$
$$\mathcal{H} = m_i(c_i T_b - L_f) + m_l(c_b T_b + w\Delta h_s) + m_h c_h T_b$$

Step 9 — Inverting $\mathcal{H}$: the regime map

Given $\mathcal{H}$, m_W, m_S, the bath state is unique: five unknowns, two mass balances, one enthalpy, and two constraints from whichever phases are present.

$$(T_b,\, m_i,\, m_l,\, m_h,\, w) = \mathcal{S}(\mathcal{H};\, m_W,\, m_S)$$

R-I — liquid only ($m_i = m_h = 0$, $m_l = m_W + m_S$, $w = \bar{w}$):

$$T_b = \frac{\mathcal{H} - m_S \Delta h_s}{(m_W + m_S)c_b}.$$

R-II — ice + brine on the liquidus ($m_h = 0$; exists only if $\bar{w} < w_E$):

$$\mathcal{H} = \Big(m_W + m_S - \frac{m_S}{w}\Big)(c_i T_{\mathrm{liq}}(w) - L_f) + \frac{m_S}{w} c_b T_{\mathrm{liq}}(w) + m_S \Delta h_s \equiv \mathcal{H}_{\mathrm{II}}(w),$$

solved for $w \in [\bar{w}, w_E]$; the root is unique since $\mathrm{d}\mathcal{H}/\mathrm{d}m_m = L_{\mathrm{eff}} > 0$ while $\mathrm{d}w/\mathrm{d}m_m < 0$. Then $T_b = T_{\mathrm{liq}}(w)$.

R-III — eutectic plateau ($T_b = T_E$, $w = w_E$):

$$\mathcal{H} = \mathcal{H}_{E0} + \Lambda_E m_l, \qquad \mathcal{H}_{E0} = m_W(c_i T_E - L_f) + m_S c_h T_E,$$

$$\Lambda_E = (1 - w_E)L_f + w_E \Delta h_s + T_E\big[c_b - (1 - w_E)c_i - w_E c_h\big],$$

$$m_l = \frac{\mathcal{H} - \mathcal{H}_{E0}}{\Lambda_E}, \qquad m_i = m_W - (1 - w_E)m_l, \qquad m_h = m_S - w_E m_l.$$

Salt runs out first (into R-II) if $\bar{w} < w_E$; ice runs out first (into R-IV) if $\bar{w} > w_E$.

R-IV — saturated brine + solid salt, no ice (only if $\bar{w} > w_E$):

$$\mathcal{H} = \frac{m_W}{1 - w_{\mathrm{sat}}}(c_b T_b + w_{\mathrm{sat}} \Delta h_s) + \Big(m_S - \frac{w_{\mathrm{sat}} m_W}{1 - w_{\mathrm{sat}}}\Big) c_h T_b \equiv \mathcal{H}_{\mathrm{IV}}(T_b),$$

solved for $T_b \in [T_E, T_\star]$, where $w_{\mathrm{sat}}(T_\star) = \bar{w}$.

R-V — fully solid ($m_l = 0$):

$$T_b = \frac{\mathcal{H} + m_W L_f}{m_W c_i + m_S c_h}.$$

Regime boundaries. The state is continuous across every boundary.

Bath type	Regime	Range	Threshold
$\bar{w} < w_E$	R-I	$\mathcal{H} \geq \mathcal{H}_A$	$\mathcal{H}_A = (m_W + m_S)c_b T_{\mathrm{liq}}(\bar{w}) + m_S \Delta h_s$
	R-II	$\mathcal{H}_B \leq \mathcal{H} < \mathcal{H}_A$	$\mathcal{H}_B = \mathcal{H}_{E0} + \Lambda_E m_S / w_E$
	R-III	$\mathcal{H}_{E0} \leq \mathcal{H} < \mathcal{H}_B$	
	R-V	$\mathcal{H} < \mathcal{H}_{E0}$	
$\bar{w} > w_E$	R-I	$\mathcal{H} \geq \mathcal{H}_C$	$\mathcal{H}_C = (m_W + m_S)c_b T_\star + m_S \Delta h_s$
	R-IV	$\mathcal{H}_D \leq \mathcal{H} < \mathcal{H}_C$	$\mathcal{H}_D = \mathcal{H}_{E0} + \Lambda_E m_W / (1 - w_E)$
	R-III	$\mathcal{H}_{E0} \leq \mathcal{H} < \mathcal{H}_D$	
	R-V	$\mathcal{H} < \mathcal{H}_{E0}$	

Result 9 — bath closure

$$T_b = T_b(\mathcal{H}), \quad m_i = m_i(\mathcal{H}), \quad m_h = m_h(\mathcal{H}), \quad w = w(\mathcal{H}) \quad \text{via R-I to R-V}$$

Step 10 — Effective latent heat, bath drift and the blank run

10.1 Differentiate along the liquidus (R-II, $m_l = m_W + m_S - m_i$):

$$\mathrm{d}\mathcal{H} = \big[h_{\mathrm{ice}}(T_b) - c_b T_b\big]\mathrm{d}m_i + \underbrace{(m_i c_i + m_l c_b)}_{C_B}\mathrm{d}T_b.$$

10.2 Per kilogram melted.

$$L_{\mathrm{eff}} \equiv \frac{\mathrm{d}\mathcal{H}}{\mathrm{d}m_m} = L_f + (c_b - c_i)T_b + C_B \frac{\mathrm{d}T_b}{\mathrm{d}m_m}.$$

10.3 Drift per kilogram melted.

$$\frac{\mathrm{d}T_b}{\mathrm{d}m_m} = -\frac{w}{m_l}\frac{\mathrm{d}T_{\mathrm{liq}}}{\mathrm{d}w} = \frac{100\, w}{m_l\,(b_1 + 2b_2\vartheta + 3b_3\vartheta^2)} > 0.$$

10.4–10.5 Effective heat capacity, every regime.

$$C_{\text{eff}} \equiv \frac{\mathrm{d}\mathcal{H}}{\mathrm{d}T_b} = \begin{cases} (m_W + m_S)c_b & \text{R-I} \\ C_B + \dfrac{L_f + (c_b - c_i)T_b}{\mathrm{d}T_b/\mathrm{d}m_m} & \text{R-II} \\ \infty & \text{R-III} \\ m_l c_b + m_h c_h + \dfrac{m_W\, w'_{\text{sat}}(T_b)}{(1 - w_{\text{sat}})^2}\left[(c_b - c_h)T_b + \Delta h_s\right] & \text{R-IV} \\ m_W c_i + m_S c_h & \text{R-V} \end{cases}$$

10.6 The plateau as a thermostat ($\dot{Q}_{\text{in}} = \dot{Q}_{db} + \dot{Q}_{\text{leak}} + P_{\text{stir}}$):

$$\frac{\mathrm{d}m_l}{\mathrm{d}t} = \frac{\dot{Q}_{\text{in}}}{\Lambda_E}, \quad \frac{\mathrm{d}m_i}{\mathrm{d}t} = -(1 - w_E)\frac{\dot{Q}_{\text{in}}}{\Lambda_E}, \quad \frac{\mathrm{d}m_h}{\mathrm{d}t} = -w_E\frac{\dot{Q}_{\text{in}}}{\Lambda_E}, \quad t_{\text{buffer}} = \frac{m_h(0)\Lambda_E}{w_E\dot{Q}_{\text{in}}}.$$

10.7 Blank-run inversion (no drink, R-II):

$$\dot{Q}_{\text{leak}} + P_{\text{stir}} = L_{\text{eff}}\frac{\mathrm{d}T_b/\mathrm{d}t}{\mathrm{d}T_b/\mathrm{d}m_m} = C_{\text{eff}}\frac{\mathrm{d}T_b}{\mathrm{d}t} \;\Rightarrow\; UA_{\text{box}} = \frac{C_{\text{eff}}\dot{T}_b - P_{\text{stir}}}{T_{\text{env}} - T_b}.$$

Using L_f in place of L_{eff} underestimates the leak.

Result 10 — explicit bath temperature equation

$$C_{\text{eff}}(T_b)\frac{\mathrm{d}T_b}{\mathrm{d}t} = UA(T_d - T_b) + UA_{\text{box}}(T_{\text{env}} - T_b) + P_{\text{stir}} \qquad (\dot{T}_b = 0 \text{ in R-III})$$

Step 11 — Initial state from premixing

11.1 Enthalpy of the ingredients.

$$\mathcal{H}_0 = m_{i0}(c_i T_{i0} - L_f) + m_{w0}c_w T_{w0} + m_S c_h T_{s0}, \qquad m_W = m_{i0} + m_{w0}.$$

11.2 Adiabatic mixing. $(T_{b0}, m_i, m_l, m_h, w)(0) = \mathcal{S}(\mathcal{H}_0; m_W, m_S)$. $\mathcal{H}_0$ is path-independent: one-shot, staged and pre-dissolved addition reach the same equilibrium, only at different rates.

11.3 Start on the eutectic iff $\mathcal{H}_{E0} \le \mathcal{H}_0 < \mathcal{H}_B$ ($\bar{w} < w_E$) or $\mathcal{H}_{E0} \le \mathcal{H}_0 < \mathcal{H}_D$ ($\bar{w} > w_E$).

11.4 Keep a salt buffer $m_h(0) \ge m_{\text{buf}}$:

$$\mathcal{H}_0 \le \mathcal{H}_{E0} + \frac{\Lambda_E(m_S - m_{\text{buf}})}{w_E}.$$

11.5 Minimum ice fraction $f = m_{i0}/m_W$, for any upper bound $\mathcal{H}_{\max}$:

$$f \ge f_{\text{crit}} = \frac{m_W c_w T_{w0} + m_S c_h T_{s0} - \mathcal{H}_{\max}}{m_W\,(c_w T_{w0} - c_i T_{i0} + L_f)}.$$

11.6 Latent share of the initial cooling.

$$\chi_{\text{lat}} = \frac{\Delta m\, L_f}{\Delta m\, L_f + m_{S,\text{dis}}\Delta h_s}.$$

Result 11 — initial conditions

$$T_d(0) = T_{d0}, \qquad \mathcal{H}(0) = \mathcal{H}_0 = m_{i0}(c_i T_{i0} - L_f) + m_{w0} c_w T_{w0} + m_S c_h T_{s0}$$

Step 12 — Parasitic terms: box leak and stirrer

$$\frac{1}{UA_{\text{box}}} = \frac{1}{h_{\text{air}} A_{\text{out}}} + \frac{\delta_{\text{box}}}{k_{\text{box}} A_{\text{box}}} + \frac{1}{h_{\text{in}} A_{\text{in}}} \approx \frac{\delta_{\text{box}}}{k_{\text{box}} A_{\text{box}}}, \qquad P_{\text{stir}} = \Gamma\omega.$$

If the lid is off (violates A5): $\dot{Q}_{\text{surf}} = h_s A_s (T_{\text{air}} - T_b) + \dot{m}_v h_{sg}$. If the drink is not fully immersed (violates A8): $C_d \dot{T}_d = -UA(T_d - T_b) + UA_{da}(T_{\text{air}} - T_d)$.

Result 12 — parasitic inputs

$$\dot{Q}_{\text{leak}} = UA_{\text{box}}(T_{\text{env}} - T_b), \qquad UA_{\text{box}} \approx \frac{k_{\text{box}} A_{\text{box}}}{\delta_{\text{box}}}, \qquad P_{\text{stir}} = \Gamma\omega$$

Step 13 — The closed model

Two first-order ODEs in the state $(T_d, \mathcal{H})$, closed by one heat-flow law and one algebraic map from $\mathcal{H}$ to the bath state.

13.1 Governing equations.

$$C_d \frac{\mathrm{d}T_d}{\mathrm{d}t} = -\dot{Q}_{db}, \qquad \frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = \dot{Q}_{db} + UA_{\text{box}}(T_{\text{env}} - T_b) + P_{\text{stir}}.$$

13.2 Heat-flow closure — choose one.

A (constant conductance): $\dot{Q}_{db} = UA(T_d - T_b), \quad \frac{1}{UA} = R_i + R_c + R_w + R_o$

B (drink-film convection): $\dot{Q}_{db} = K_i A_i \left[(T_d - T_b) - \dot{Q}_{db}(R_c + R_w + R_o)\right]^{5/4}, \quad K_i = C_i k_d \left(\frac{g\beta}{\nu\alpha H}\right)^{1/4}$

13.3 Bath closure. $(T_b, m_i, m_l, m_h, w) = \mathcal{S}(\mathcal{H}; m_W, m_S)$:

$$\mathcal{S}: \begin{cases} \text{R-I:} & m_i = m_h = 0,\ w = \bar{w},\ T_b = \dfrac{\mathcal{H} - m_S \Delta h_s}{(m_W + m_S) c_b} \\ \text{R-II:} & m_h = 0,\ \mathcal{H} = \left(m_W + m_S - \dfrac{m_S}{w}\right)(c_i T_{\text{liq}}(w) - L_f) + \dfrac{m_S}{w} c_b T_{\text{liq}}(w) + m_S \Delta h_s,\ T_b = T_{\text{liq}}(w) \\ \text{R-III:} & T_b = T_E,\ w = w_E,\ m_l = \dfrac{\mathcal{H} - \mathcal{H}_{E0}}{\Lambda_E},\ m_i = m_W - (1 - w_E) m_l,\ m_h = m_S - w_E m_l \\ \text{R-IV:} & m_i = 0,\ w = w_{\text{sat}}(T_b),\ \mathcal{H} = \dfrac{m_W(c_b T_b + w\Delta h_s)}{1 - w} + \left(m_S - \dfrac{w m_W}{1 - w}\right) c_h T_b \\ \text{R-V:} & m_l = 0,\ T_b = \dfrac{\mathcal{H} + m_W L_f}{m_W c_i + m_S c_h} \end{cases}$$

with the regime chosen by the Step 9 threshold table, $W = b_1\vartheta + b_2\vartheta^2 + b_3\vartheta^3$ ($W = 100w$, $\vartheta = -T_{\text{liq}}$), and

$$\mathcal{H}_{E0} = m_W(c_i T_E - L_f) + m_S c_h T_E, \quad \Lambda_E = (1 - w_E) L_f + w_E \Delta h_s + T_E\left[c_b - (1 - w_E) c_i - w_E c_h\right],$$
$$\bar{w} = \frac{m_S}{m_W + m_S}.$$

13.4 Equivalent explicit form (for Desmos / Mathematica; integrate the $\mathcal{H}$ form, which carries the plateau without dividing by $C_{\text{eff}} = \infty$):

$$C_d \dot{T}_d = -\dot{Q}_{db}, \qquad C_{\text{eff}}(T_b)\,\dot{T}_b = \dot{Q}_{db} + UA_{\text{box}}(T_{\text{env}} - T_b) + P_{\text{stir}} \quad (\dot{T}_b = 0 \text{ in R-III}).$$

13.5 Initial conditions.

$$T_d(0) = T_{d0}, \qquad \mathcal{H}(0) = m_{i0}(c_i T_{i0} - L_f) + m_{w0} c_w T_{w0} + m_S c_h T_{s0}, \qquad m_W = m_{i0} + m_{w0}.$$

13.6 Stop and validity conditions.

- T_d reaches the target T^* — the measured output.
- T_d reaches the drink's freezing point T_{fd} — beyond this the drink needs its own enthalpy model.
- In a buffered run, $m_h \to 0$ — the thermostat is gone and the bath leaves R-III.
- T_d approaches the density maximum — closure B and A1 weaken.

13.7 Parameters and where each comes from.

Group	Parameters	How fixed
Drink	$C_d = m_d c_d + m_c c_c$	weighing + tables
Heat path	UA (A) or $K_i A_i, R_c, R_w, R_o$ (B)	fitted on a buffered run, where $T_b = T_E$ is known
Bath composition	m_W, m_S, m_{i0}, m_{w0}	weighing
Bath properties	$c_i, c_b, c_h, c_w, L_f, \Delta h_s$	tables
Phase diagram	$b_1, b_2, b_3, T_E, w_E, w_{\text{sat}}(T)$	Bodnar, literature, own coexistence curve
Environment	$UA_{\text{box}}, P_{\text{stir}}, T_{\text{env}}(t)$	blank runs, logged channel 4
Start	$T_{d0}, T_{i0}, T_{w0}, T_{s0}$	logged

Once UA is fixed on a buffered run, an unbuffered run has no free parameter left; it is a prediction.

Result 13 — the current model, in one block

$$\begin{aligned} C_d\,\dot{T}_d &= -\,UA\,(T_d - T_b) \\ \dot{\mathcal{H}} &= UA\,(T_d - T_b) + UA_{\text{box}}(T_{\text{env}} - T_b) + P_{\text{stir}} \\ T_b &= T_b(\mathcal{H};\, m_W,\, m_S) \;\text{ via } \mathcal{S} \\ T_d(0) &= T_{d0}, \qquad \mathcal{H}(0) = \mathcal{H}_0 \end{aligned}$$

Step 14 — Closed-form special cases

14.1 Buffered bath (R-III throughout, $T_b = T_E$): Result 4 with $T_b = T_E$, valid while $t < t_{\text{buffer}}$ (10.6):

$$T_d(t) = T_E + (T_{d0} - T_E)\,e^{-t/\tau}.$$

14.2 Linearised liquidus bath (R-II, C_{eff} constant, $UA_{\text{box}} = P_{\text{stir}} = 0$): a two-capacitor problem. Subtracting the two equations of 13.4,

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -UA\Big(\frac{1}{C_d} + \frac{1}{C_{\text{eff}}}\Big)\theta \;\Rightarrow\; \theta = \theta_0 e^{-t/\tau_c}, \qquad \tau_c = \frac{C_d C_{\text{eff}}}{UA\,(C_d + C_{\text{eff}})} = \frac{\tau}{1+\Gamma}, \quad \Gamma = \frac{C_d}{C_{\text{eff}}}.$$

Energy conservation, $C_d T_d + C_{\text{eff}} T_b = \text{const}$, fixes the shared final temperature:

$$T_\infty = \frac{C_d T_{d0} + C_{\text{eff}} T_{b0}}{C_d + C_{\text{eff}}}, \qquad T_d(t) = T_\infty + (T_{d0} - T_\infty)e^{-t/\tau_c}, \qquad T_b(t) = T_\infty - (T_\infty - T_{b0})e^{-t/\tau_c}.$$

The drink reaches T^* only if $T_\infty < T^*$, after $t_{T^*} = \tau_c \ln\big[(T_{d0} - T_\infty)/(T^* - T_\infty)\big]$.

14.3 With the box leak (T_{env} constant): a linear system $\dot{\mathbf{x}} = \mathsf{A}\mathbf{x} + \mathbf{f}$, $\mathbf{x} = (T_d, T_b)^{\mathsf{T}}$,

$$\mathsf{A} = \begin{pmatrix} -\dfrac{UA}{C_d} & \dfrac{UA}{C_d} \\[2ex] \dfrac{UA}{C_{\text{eff}}} & -\dfrac{UA + UA_{\text{box}}}{C_{\text{eff}}} \end{pmatrix}, \qquad \lambda_\pm = \tfrac{1}{2}\Big[\operatorname{tr}\mathsf{A} \pm \sqrt{(\operatorname{tr}\mathsf{A})^2 - 4\det\mathsf{A}}\Big],$$

$$\operatorname{tr}\mathsf{A} = -\Big(\frac{UA}{C_d} + \frac{UA + UA_{\text{box}}}{C_{\text{eff}}}\Big), \qquad \det\mathsf{A} = \frac{UA \cdot UA_{\text{box}}}{C_d\, C_{\text{eff}}}.$$

Both eigenvalues are negative; the slow one is set by the leak, the fast one by UA.

14.4 Fixed bath, convection-limited drink: Result 5, $\theta = \theta_0(1 + t/t^*)^{-4}$, $t^* = 4\tau_0$.